Concise Data Dictionary — Review 2

Project: Laboratory Equipment Booking & Maintenance System

This data dictionary defines the structure, data types, nullability, primary/foreign key relationships, and integrity constraints for all 12 relational tables normalized up to Third Normal Form (3NF), followed by the 3 triggers that enforce the booking and lifecycle rules.

Target DBMS: PostgreSQL 15+.

1. Laboratory Table

Description: Physical laboratories that house equipment.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
laboratory_id	INTEGER	PK	NOT NULL	Unique laboratory identifier.
name	VARCHAR(100)	UNIQUE	NOT NULL	Laboratory name.
location	VARCHAR(100)	-	NOT NULL	Block and floor of the laboratory.

2. EquipmentCategory Table

Description: Catalog of equipment categories.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
category_id	INTEGER	PK	NOT NULL	Unique category identifier.
name	VARCHAR(80)	UNIQUE	NOT NULL	Category name.
description	VARCHAR(200)	-	NOT NULL	What the category covers.

3. Vendor Table

Description: Service and calibration vendors used for maintenance.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
vendor_id	INTEGER	PK	NOT NULL	Unique vendor identifier.
name	VARCHAR(100)	UNIQUE	NOT NULL	Vendor name.
phone	VARCHAR(15)	-	NOT NULL	Contact number.
email	VARCHAR(100)	-	NOT NULL	CHECK (email LIKE '%_@_%._%') basic email format.

4. AppUser Table

Description: Requesters, lab staff and administrators (ER entity USER; renamed because USER is reserved in PostgreSQL).

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
user_id	INTEGER	PK	NOT NULL	Unique user identifier.
name	VARCHAR(80)	-	NOT NULL	User full name.
email	VARCHAR(100)	UNIQUE	NOT NULL	CHECK (email LIKE '%_@_%._%') basic email format.
role	VARCHAR(12)	-	NOT NULL	CHECK (role IN ('STUDENT', 'LAB_STAFF', 'ADMIN')).

5. Equipment Table

Description: Master catalog of laboratory equipment; status decides whether it can be booked.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
equipment_id	INTEGER	PK	NOT NULL	Unique equipment identifier.
laboratory_id	INTEGER	FK	NOT NULL	References Laboratory(laboratory_id).
category_id	INTEGER	FK	NOT NULL	References EquipmentCategory(category_id).
name	VARCHAR(100)	-	NOT NULL	Equipment name.
serial_no	VARCHAR(30)	UNIQUE	NOT NULL	Manufacturer serial number.
purchase_date	DATE	-	NOT NULL	Date of purchase.
status	VARCHAR(20)	-	NOT NULL	DEFAULT 'ACTIVE'. CHECK (status IN ('ACTIVE', 'UNDER_MAINTENANCE', 'BROKEN', 'RETIRED')). Only ACTIVE equipment can be booked.

6. TimeSlot Table

Description: Bookable time windows on a given date.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
slot_id	INTEGER	PK	NOT NULL	Unique slot identifier.
slot_date	DATE	-	NOT NULL	Date of the slot.
start_time	TIME	-	NOT NULL	Slot start time.
end_time	TIME	-	NOT NULL	CHECK (end_time > start_time). Slot end time.

Table constraint: Unique (slot_date, start_time, end_time).

7. Booking Table

Description: Equipment reservation requests and their approval status.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
booking_id	INTEGER	PK	NOT NULL	Unique booking identifier.
user_id	INTEGER	FK	NOT NULL	References AppUser(user_id).
equipment_id	INTEGER	FK	NOT NULL	References Equipment(equipment_id).
slot_id	INTEGER	FK	NOT NULL	References TimeSlot(slot_id).
status	VARCHAR(12)	-	NOT NULL	DEFAULT 'PENDING'. CHECK (status IN ('PENDING', 'APPROVED', 'REJECTED', 'CANCELLED', 'COMPLETED')).
requested_on	TIMESTAMP	-	NOT NULL	DEFAULT CURRENT_TIMESTAMP. When the request was made.
purpose	VARCHAR(150)	-	NOT NULL	Reason for the booking.

Index: idx_booking_equipment_status ON Booking(equipment_id, status). Trigger: trg_validate_booking (see Section 13).

8. UsageLog Table

Description: Actual usage recorded against a booking.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
usage_log_id	INTEGER	PK	NOT NULL	Unique usage log identifier.
booking_id	INTEGER	FK	NOT NULL	References Booking(booking_id) ON DELETE CASCADE.
usage_date	DATE	-	NOT NULL	Date the equipment was used.
hours_used	NUMERIC(4,2)	-	NOT NULL	CHECK (hours_used > 0 AND hours_used <= 12) hours of use.
details	VARCHAR(200)	-	NOT NULL	Description of the work done.

9. Calibration Table

Description: Calibration history and next due dates for each equipment.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
calibration_id	INTEGER	PK	NOT NULL	Unique calibration identifier.
equipment_id	INTEGER	FK	NOT NULL	References Equipment(equipment_id) ON DELETE CASCADE.
calibration_date	DATE	-	NOT NULL	Date the calibration was performed.
next_due_date	DATE	-	NOT NULL	CHECK (next_due_date > calibration_date). Next calibration due date.
performed_by	VARCHAR(100)	-	NOT NULL	Person or agency that calibrated the equipment.

Table constraint: Unique (equipment_id, calibration_date).

10. Breakdown Table

Description: Reported equipment failures and their resolution.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
breakdown_id	INTEGER	PK	NOT NULL	Unique breakdown identifier.
equipment_id	INTEGER	FK	NOT NULL	References Equipment(equipment_id) ON DELETE CASCADE.
reported_on	DATE	-	NOT NULL	Date the failure was reported.
resolved_on	DATE	-	NULL	CHECK (resolved_on IS NULL OR resolved_on >= reported_on).
description	VARCHAR(200)	-	NOT NULL	Fault description.
status	VARCHAR(12)	-	NOT NULL	DEFAULT 'REPORTED'. CHECK (status IN ('REPORTED', 'IN_REPAIR', 'RESOLVED')). RESOLVED if and only if resolved_on is not NULL.

Table constraint: Unique (breakdown_id, equipment_id), the target of the composite FK from MaintenanceJob. Trigger: trg_breakdown_reported.

11. MaintenanceJob Table

Description: Service jobs performed by vendors, optionally linked to a breakdown.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
maintenance_id	INTEGER	PK	NOT NULL	Unique maintenance job identifier.
equipment_id	INTEGER	FK	NOT NULL	References Equipment(equipment_id) ON DELETE CASCADE. Part of composite FK (breakdown_id, equipment_id).
vendor_id	INTEGER	FK	NOT NULL	References Vendor(vendor_id).
breakdown_id	INTEGER	FK	NULL	Composite FK (breakdown_id, equipment_id) references Breakdown(breakdown_id, equipment_id); NULL for preventive jobs.
description	VARCHAR(200)	-	NOT NULL	Work performed or planned.
start_date	DATE	-	NOT NULL	Job start date.
completion_date	DATE	-	NULL	CHECK (completion_date IS NULL OR completion_date >= start_date).
status	VARCHAR(12)	-	NOT NULL	DEFAULT 'SCHEDULED'. CHECK (status IN ('SCHEDULED', 'IN_PROGRESS', 'COMPLETED')). COMPLETED if and only if completion_date is not NULL.
cost	NUMERIC(10,2)	-	NOT NULL	DEFAULT 0. CHECK (cost >= 0) maintenance cost in INR.

Trigger: trg_maintenance_lifecycle (see Section 13).

12. Consumable Table

Description: Consumable items attached to equipment, with stock control.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
consumable_id	INTEGER	PK	NOT NULL	Unique consumable identifier.
equipment_id	INTEGER	FK	NOT NULL	References Equipment(equipment_id) ON DELETE CASCADE.
item_name	VARCHAR(100)	-	NOT NULL	Consumable item name.
unit	VARCHAR(15)	-	NOT NULL	Unit of measure (pcs, bottles, sheets).
stock_quantity	INTEGER	-	NOT NULL	CHECK (stock_quantity >= 0) stock can never go below zero.
reorder_level	INTEGER	-	NOT NULL	CHECK (reorder_level >= 0) quantity at which restocking is needed.

Table constraint: Unique (equipment_id, item_name).

13. Triggers & Business Rules

Description: Rules a CHECK constraint cannot enforce because they compare separate rows or tables.

Trigger	Fires On	Rule Enforced
trg_validate_booking	Booking (BEFORE INSERT OR UPDATE OF equipment_id, slot_id, status)	For PENDING/APPROVED rows: locks the equipment row, rejects non-ACTIVE equipment, rejects a booking whose latest calibration expires before the slot date, and for APPROVED rows rejects overlap with another approved booking of the same equipment (existing.start < requested.end AND existing.end > requested.start).
trg_breakdown_reported	Breakdown (AFTER INSERT, status not RESOLVED)	Sets an ACTIVE equipment to BROKEN.
trg_maintenance_lifecycle	MaintenanceJob (AFTER INSERT OR UPDATE OF status)	IN_PROGRESS: equipment becomes UNDER_MAINTENANCE and the breakdown IN_REPAIR. COMPLETED: breakdown becomes RESOLVED (resolved_on = completion_date) and equipment returns to ACTIVE when no unresolved breakdown remains.